

Cover letter — Results in Engineering

Dear Editors,

We submit the enclosed manuscript, “A pre-deployment three-way triage gate for visual-inspection anomaly detectors: finite-sample escaped-defect and false-reject certificates on academic AD benchmarks,” for consideration as a research article in *Results in Engineering*.

Anomaly detectors can rank defects well without defining a safe operating decision. This paper presents a pre-deployment triage layer that converts anomaly scores into per-category auto-pass, auto-reject, and defer regions with finite-sample split-conformal bounds on escaped-defect and false-reject rates. Categories whose calibration pools cannot support a requested bound are reported as audited-not-certified rather than silently loosened. Results on MVTec AD, VisA, and MPDD show when dual certification holds, when refusal dominates, and that the evidence supports a calibration/refusal protocol on academic AD benchmarks, not portable production-line deployment.

This manuscript is original and is not under consideration elsewhere. The author declares no competing interests. Corresponding author: Zeyu Fu (fuzeyu09@gmail.com).

Thank you for your consideration.

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